

Integration by Parts, Chosen Strategically

Answer Key

AP Calculus AB/BC

Quick Answers

1. $xe^x - e^x$

2. $x \sin(x) + \cos(x)$

3. $x \log(x) - x$

4. $(x^2 - 2x + 2)e^x$

5. $\frac{1}{4} + \frac{e^2}{4}$

6. $\frac{x^2 \log(x)}{2} - \frac{x^2}{4}$

7. $x \tan(x) + \log(|\cos(x)|)$

8. $\frac{(\sin(x) - \cos(x))e^x}{2}$

9. -2π

10. 0, 1

Curriculum

Level: AP Calculus AB/BC

FUN-6.E – problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Difficulty 1–5 of 5 across 11 tagged checks.

1. $\int xe^x dx$. LIATE puts the Algebraic factor ahead of the Exponential, so $u = x$ and $dv = e^x dx$. That choice is the whole point: differentiating x kills it, while e^x integrates to itself.

$$u = x, \quad du = dx, \quad dv = e^x dx, \quad v = e^x$$
$$\int xe^x dx = xe^x - \int e^x dx$$

The remaining integral is elementary. $\boxed{xe^x - e^x + C}$

2. $\int x \cos x dx$. Same reasoning: $u = x$ (Algebraic) and $dv = \cos x dx$ (Trigonometric).

$$u = x, \quad du = dx, \quad dv = \cos x dx, \quad v = \sin x$$
$$\int x \cos x dx = x \sin x - \int \sin x dx = x \sin x + \cos x + C$$

Check by differentiating: $\sin x + x \cos x - \sin x = x \cos x$. $\boxed{x \sin x + \cos x + C}$

3. $\int \ln x \, dx$. There is only one factor, so write the integrand as $\ln x \cdot 1$. The Logarithm is first in LIATE, which forces $u = \ln x$ and $dv = dx$ — and that is the only workable split, since $\ln x$ has no elementary antiderivative to serve as v .

$$u = \ln x, \quad du = \frac{1}{x} dx, \quad dv = dx, \quad v = x$$

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int dx$$

$$\boxed{x \ln x - x + C}$$

4. $\int x^2 e^x \, dx$. With $u = x^2$, one round of parts leaves $\int x e^x \, dx$ — not finished, but strictly simpler, because the power dropped by one. That is the signal to apply parts again rather than to change method.

$$\int x^2 e^x \, dx = x^2 e^x - 2 \int x e^x \, dx$$

and problem 1 already evaluated the inner integral:

$$= x^2 e^x - 2(xe^x - e^x)$$

$$\boxed{e^x (x^2 - 2x + 2) + C}$$

5. $\int_0^1 x e^{2x} \, dx$. Parts first, limits last. With $u = x$ and $dv = e^{2x} \, dx$, the chain rule runs backwards: $v = \frac{1}{2} e^{2x}$, and forgetting that $\frac{1}{2}$ is the most common slip here.

$$\int x e^{2x} \, dx = \frac{x}{2} e^{2x} - \frac{1}{2} \int e^{2x} \, dx = \frac{x}{2} e^{2x} - \frac{1}{4} e^{2x}$$

$$\left[\right]_0^1 = \left(\frac{e^2}{2} - \frac{e^2}{4} \right) - \left(0 - \frac{1}{4} \right)$$

$$\boxed{\frac{e^2 + 1}{4}}$$

6. $\int x \ln x \, dx$. Why $u = \ln x$ and not $u = x$: taking $u = x$ would need $v = \int \ln x \, dx$, which is itself a parts problem, so the integral would get harder. Taking $u = \ln x$ replaces the logarithm by $\frac{1}{x}$ and leaves a power to integrate.

$$u = \ln x, \quad du = \frac{1}{x} dx, \quad dv = x \, dx, \quad v = \frac{x^2}{2}$$

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} dx = \frac{x^2}{2} \ln x - \int \frac{x}{2} dx$$

$$\boxed{\frac{x^2}{2} \ln x - \frac{x^2}{4} + C}$$

7. $\int x \sec^2 x \, dx$. The strategic move is to notice that $\sec^2 x$ is the factor whose antiderivative you already know, so it must be dv ; $u = x$ then differentiates away.

$$u = x, \quad du = dx, \quad dv = \sec^2 x \, dx, \quad v = \tan x$$

$$\int x \sec^2 x \, dx = x \tan x - \int \tan x \, dx$$

The leftover integral is a standard one: $\int \tan x \, dx = -\ln |\cos x|$, so subtracting it adds the logarithm back. $\boxed{x \tan x + \ln |\cos x| + C}$

8. $\int e^x \sin x \, dx$. Neither factor simplifies under differentiation, so parts cannot terminate — it cycles. Call the integral I , apply parts twice keeping the same kind of choice both times, and an equation in I appears.

$$I = \int e^x \sin x \, dx = e^x \sin x - \int e^x \cos x \, dx$$

$$\int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx = e^x \cos x + I$$

$$I = e^x \sin x - e^x \cos x - I$$

Solving the equation, $2I = e^x(\sin x - \cos x)$. $\boxed{\frac{e^x(\sin x - \cos x)}{2} + C}$

9. $\int_0^\pi x^2 \cos x \, dx$. The power is 2, so expect two rounds of parts, each with u the polynomial.

$$\begin{aligned}\int x^2 \cos x \, dx &= x^2 \sin x - 2 \int x \sin x \, dx \\ \int x \sin x \, dx &= -x \cos x + \sin x \\ \Rightarrow \int x^2 \cos x \, dx &= x^2 \sin x + 2x \cos x - 2 \sin x\end{aligned}$$

At $x = \pi$: $\sin \pi = 0$ and $\cos \pi = -1$, giving $0 + 2\pi(-1) - 0$. At $x = 0$ every term is zero, so the lower limit contributes nothing. $\boxed{-2\pi}$

The sign is the thing to check: a positive answer means the minus sign in the second round of parts was dropped.

10. $\int_1^\infty \frac{\ln x}{x^2} \, dx$. Never substitute ∞ into an antiderivative — integrate to a finite t , then take a limit. Parts with $u = \ln x$, $dv = x^{-2}dx$, $v = -\frac{1}{x}$:

$$\begin{aligned}\int_1^t \frac{\ln x}{x^2} \, dx &= \left[-\frac{\ln x}{x} \right]_1^t + \int_1^t \frac{1}{x^2} \, dx = -\frac{\ln t}{t} + 0 + \left[-\frac{1}{x} \right]_1^t \\ &= -\frac{\ln t}{t} - \frac{1}{t} + 1 = -\frac{\ln t + 1}{t} + 1\end{aligned}$$

(a) As $t \rightarrow \infty$ the denominator grows linearly while the numerator grows logarithmically, so the quotient collapses; L'Hôpital's rule on $\frac{\ln t + 1}{t}$ gives $\frac{1/t}{1} \rightarrow 0$. Polynomial growth beats logarithmic growth. $\boxed{0}$

(b) Only the constant term survives the limit, so the integral converges. $\boxed{1}$